Modern Application Development-1 Project Report: Placement Portal Application

1. Student Details

Name: Sibani Tiwari
Roll number: 23F3003311
Program: BS in Data Science and Applications
Institution: Indian Institute of Technology Madras
Course: Modern Application Development – 1
Project Title: Placement Portal Application
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2. Project Details

Project Title: Placement Portal Application

Problem Statement:
Institutes require an efficient system to manage campus recruitment activities involving students, companies, and placement drives. Manual processes such as spreadsheets and emails often lead to poor tracking, duplicate applications, approval delays, and inconsistent record management.

This project aims to build a role-based web application using Flask that enables structured interaction between Admin (Institute), Company, and Student. The system ensures secure authentication, role-based access control, and automatic database creation using SQLite.

Approach:
The application was built using Flask with a modular structure separating models, routes, and templates. The system implements three roles:

- **Admin:** Approves companies and drives, manages users, views statistics, and maintains complete placement records.
- **Company:** Registers (subject to approval), creates placement drives, and updates student application status.
- **Student:** Registers, views approved drives, applies, and tracks application history.

Relational database design was used to prevent redundancy and maintain data integrity. Validation logic prevents duplicate applications and ensures only approved companies can create drives.

3. AI/LLM Declaration

Component / Tool	Purpose in Project	Usage %
Flask (App + Routing)	Backend logic, routing, request handling, role control	22%
SQLite + SQLAlchemy	Data modeling, relationships, database persistence	18%
HTML + Jinja2	Server-side template rendering (dashboards, forms, tables)	20%
Bootstrap / CSS	UI styling, layout, responsiveness	8%
Flask-Login (Authentication)	Login/logout, role-based access (Admin/Company/Student)	10%
Admin CRUD Logic	Approve/reject drives, manage companies, blacklist users	8%
Student & Company Logic	Drive creation, application flow, status updates	8%
Validation (Backend + HTML5)	Input validation, duplicate prevention, constraints	3%
ChartJS (Visualization)	Placement statistics & analytics (optional feature)	2%
REST APIs (JSON Endpoints)	/api/drives, /api/admin/applications, stats endpoints	1%

Category wise tools Summary:

Category	Total %
Backend (Flask + Database)	40%
Frontend (Jinja + HTML)	20%
Authentication & Access Control	10%
Business Logic (Admin + User Flow)	16%
UI / Styling	8%
Optional Enhancements (Charts + APIs)	6%

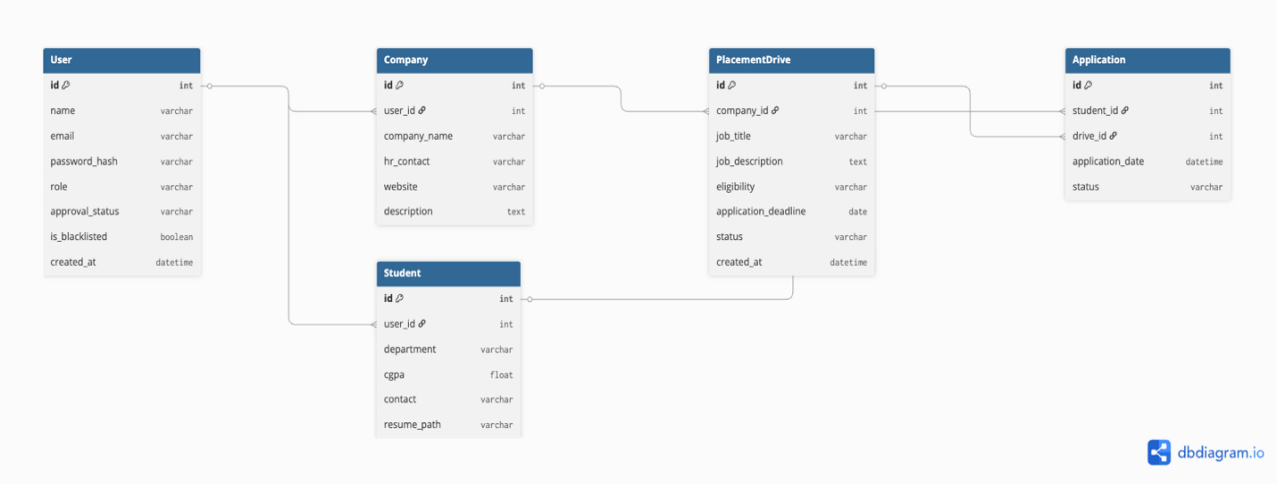
4. Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Core backend web framework
Flask-Login	Authentication and role-based access control

Technology / Library	Purpose
SQLAlchemy	ORM for SQLite database
SQLite	Local database
Jinja2	Template rendering
HTML + CSS	Frontend structure
Werkzeug Security	Password hashing
Chart.js (Optional)	Placement statistics visualization
JSON APIs	Optional resource endpoints

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5. Database Schema / ER Diagram



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6. API Resource Endpoints

Endpoint	Method	Description
/api/drives	GET	Fetch approved placement drives
/api/admin/applications	GET	Fetch all applications (Admin only)

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7. Architecture Overview

- app.py – Main Flask entry
- models.py – SQLAlchemy database models
- routes.py – Business logic and routing
- templates/ – Jinja2 HTML templates
- static/ - resumes and assets

Implemented Features:

- Role-based login system
- Company approval workflow
- Placement drive creation and management
- Student application tracking
- Duplicate prevention logic
- Resume upload support
- Admin statistics dashboard

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8. Video Presentation

Drive Link:

https://drive.google.com/file/d/1USwptyjeV4m8f7gykT5XQ0qComHurzdn/view?usp=sharing

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